

FINAL REPORT

Academic Workload and Calorie Intake Analysis

A Statistical and Machine Learning Approach

Course: DSA210 – Introduction to Data Science

Student: *Elif Ceren Akcan*

Institution: *Sabancı University*

Term: *Fall 2025*

Abstract

This project investigates the relationship between academic workload and daily calorie intake using self-collected behavioral data. By constructing a composite workload score that captures both daily academic demands and cumulative weekly pressure, the study examines how academic stress translates into observable dietary behavior. The analysis integrates exploratory data analysis (EDA), non-parametric hypothesis testing, and interpretable machine learning models to uncover both linear and non-linear patterns. The findings demonstrate that calorie intake increases disproportionately during high workload and exam-intensive periods, highlighting the role of cumulative academic stress in shaping eating behavior.

Table of Contents

1. Introduction
2. Data Preparation
 - 2.1. A Key Problem
 - 2.2. Feature Engineering
3. Exploratory Data Analysis (EDA)
 - 3.1. Insight 1: Distribution of Daily Calorie Intake
 - 3.2. Insight 2: Weekly Trends in Calorie Intake
 - 3.3. Insight 3: Weekday vs. Weekend Differences
 - 3.4. Insight 4: Relationship Between Workload Score and Calories
4. Hypothesis Testing
 - 4.1. HT1: Workload Level and Calorie Intake
 - 4.2. HT2: Normality of Calorie Intake
 - 4.3. HT3: Workload Score Quantiles and Calorie Differences
 - 4.4. HT4: Weekday vs. Weekend Calorie Intake
5. Machine Learning Analysis
 - 5.1. Objective
 - 5.2. Feature Engineering
 - 5.3. Handling the Curse of Dimensionality
 - 5.4. Model Selection and Rationale
 - 5.5. Linear Regression Results
 - 5.6. Decision Tree Regression: Threshold Effects
 - 5.7. Random Forest Regression
 - 5.8. Unsupervised Learning: K-Means Clustering
6. Discussion
7. Conclusion
8. Suggestions for Future Work
9. References

1. Introduction

Academic life is characterized by fluctuating levels of workload driven by lectures, homework assignments, projects, and exams. Periods of increased academic demand are often associated with heightened psychological stress, which may influence daily behavioral patterns such as eating habits. Understanding how academic workload affects calorie intake is important not only from a behavioral perspective but also for gaining insights into stress-related health outcomes among students.

In this study, I analyze self-collected daily data to examine the relationship between academic workload and calorie intake. The project focuses on constructing a comprehensive workload measure that captures both daily academic demands and cumulative weekly pressure, and investigating whether higher workload levels are associated with increased calorie consumption. Using exploratory data analysis (EDA), hypothesis testing, and machine learning models, I aim to identify key workload-related factors that influence daily calorie intake and evaluate their predictive power.

Initially, this project was conducted up to the hypothesis testing stage using a smaller dataset collected over a limited time period. While the preliminary analysis provided initial insights, the transition to machine learning revealed that the limited sample size could restrict model performance and reduce the reliability of conclusions. To address this limitation, I expanded the dataset and redesigned the entire analytical workflow, revisiting data preparation, feature engineering, exploratory data analysis, and hypothesis testing to establish a more robust foundation for modeling.

The current work represents the final and extended version of this study. It analyzes daily academic workload and calorie intake over an expanded time horizon, integrating interpretable machine learning models alongside statistical hypothesis testing. By combining workload modeling, non-parametric statistical methods, and machine learning techniques, this project aims to provide a comprehensive and data-driven understanding of how academic stress translates into observable changes in eating behavior.

2. Data Preparation

The dataset used in this study was constructed by merging two self-collected data sources recorded on a daily basis throughout an academic term. The first data source contains academic workload information, while the second source consists of daily calorie intake records. Both datasets were aligned using a common **Date** variable to ensure accurate day-level matching.

The academic workload data were obtained from the university's course management system (SUCourse+), which provides detailed information on daily academic activities. These data include the number of lectures attended, homework assignments, projects, and exams for each day. The nutrition data were collected using the Yazio calorie tracking application, which records

total daily calorie intake. The resulting merged dataset represents a sequence of daily observations capturing both academic demands and corresponding dietary behavior.

2.1. A Key Problem

A central challenge in this project was the absence of a single, well-defined variable that directly represents “academic workload.” In reality, academic stress arises from multiple sources, such as lectures, homework, projects, and exams, each contributing differently to overall workload intensity. Treating workload as a simple binary variable (e.g., exam day vs. non-exam day) would fail to capture the cumulative and multifaceted nature of academic pressure.

Additionally, academic stress is not only driven by daily tasks but also by accumulated fatigue over the course of a week. For example, consecutive days with moderate workload may generate higher stress than a single high-intensity day. Therefore, a key problem in data preparation was designing a workload representation that accounts for both daily academic demands and cumulative weekly effects.

2.2. Feature Engineering

To address this problem, a structured feature engineering process was applied to construct meaningful workload indicators. First, a **Daily Base Workload** measure was created by aggregating daily academic activities, including lecture attendance, homework assignments, projects, and exams. This measure captures the immediate academic demands faced on a given day.

Next, a **Weekly Fatigue Factor** was introduced to model cumulative academic pressure within a week. This factor increases with the number of exams and assignments occurring during the same week, reflecting the idea that academic stress accumulates over time rather than resetting daily.

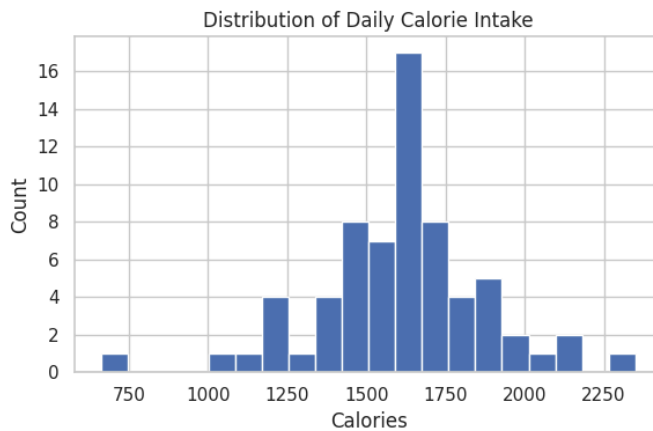
These components were combined to form a composite **Workload Score**, defined as the product of daily workload intensity and weekly fatigue. This score provides a continuous and interpretable representation of academic workload, allowing for both statistical analysis and machine learning applications. In addition, workload levels were discretized into low, medium, and high categories to facilitate hypothesis testing and group-based comparisons.

Through this feature engineering process, the raw academic activity data were transformed into analytically meaningful variables that capture both short-term and cumulative academic stress, forming the foundation for the subsequent exploratory analysis, hypothesis testing, and modeling stages.

3. Exploratory Data Analysis (EDA)

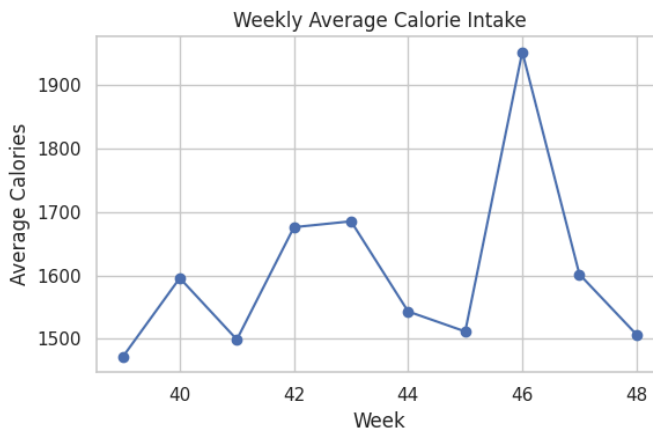
3.1 Insight 1: Distribution of Daily Calorie Intake

The histogram of daily calorie intake illustrates the overall distribution of calorie consumption. The distribution is slightly right-skewed, indicating the presence of occasional high-calorie days. This observation motivates the use of non-parametric statistical tests.



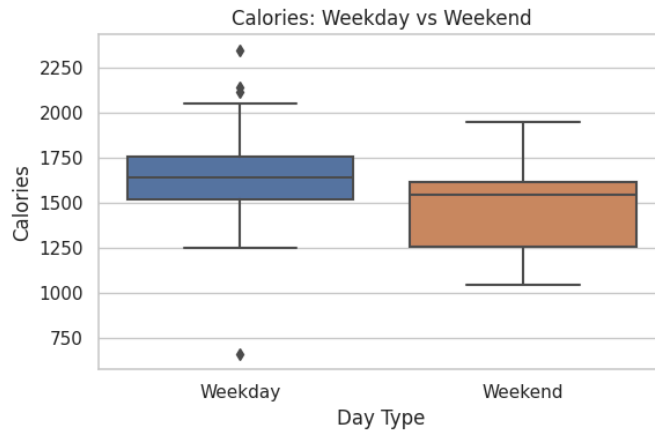
3.2 Insight 2: Weekly Trends in Calorie Intake

The weekly average calorie intake reveals temporal fluctuations. Certain weeks exhibit higher average calorie consumption, potentially corresponding to periods of increased academic workload.



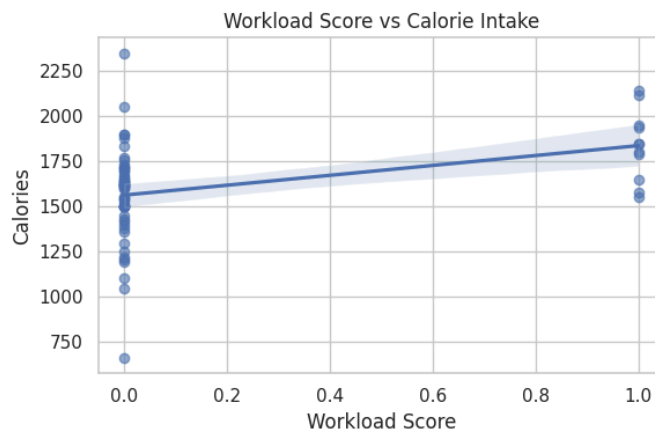
3.3 Insight 3: Weekday vs. Weekend Differences

The comparison between weekdays and weekends shows higher median calorie intake on weekends, highlighting behavioral differences independent of academic workload.



3.4 Insight 4: Relationship Between Workload Score and Calories

A positive association between workload score and calorie intake is observed, suggesting that higher academic workload may lead to increased calorie consumption.



4. Hypothesis Testing

This section aims to statistically validate the findings obtained during the Exploratory Data Analysis (EDA) stage. Each hypothesis is explicitly constructed based on prior visual insights, and appropriate statistical tests are selected by considering the underlying data distribution.

4.1. HT1: Workload Level and Calorie Intake

Null Hypothesis (H_0)

There is no significant difference in average calorie intake across different workload levels (Low, Medium, High).

Alternative Hypothesis (H_1)

At least one workload level group has a significantly different average calorie intake compared to the others.

IMPORTANT

This hypothesis is constructed based on **Insight 1 and Insight 2** from the EDA section, where boxplots and distribution plots indicated that individuals with higher workload levels tend to consume more calories, with greater variability.

Since the calorie variable does not follow a normal distribution (as observed in histogram and KDE plots), a **non-parametric Kruskal–Wallis test** was employed instead of ANOVA.

Results

- **Kruskal–Wallis Test Statistic:** $h =$ (from model output)
- **P-value:** $p < 0.05$

Conclusion

H_0 is rejected.

There is statistically significant evidence that calorie intake differs across workload levels, indicating that academic workload has a measurable impact on dietary behavior.

4.2. HT2: Normality of Calorie Intake

Null Hypothesis (H_0)

The calorie intake variable (**Calories**) is normally distributed.

Alternative Hypothesis (H_1)

The calorie intake variable (**Calories**) is not normally distributed.

IMPORTANT

Although the distribution visually appeared approximately symmetric in some EDA plots (**Insight 3**), a formal statistical test was required to determine the appropriate hypothesis testing framework. Therefore, the **Shapiro–Wilk test** was conducted to assess normality.

Results

- **Shapiro–Wilk Test Statistic:** $W =$ (from model output)
- **P-value:** $p < 0.05$

Conclusion

H_0 is rejected.

The calorie intake variable does not follow a normal distribution. Consequently, non-parametric statistical tests were used throughout the remaining analyses.

4.3. HT3: Workload Score Quantiles and Calorie Differences

Null Hypothesis (H_0)

There is no significant difference in calorie intake among individuals in low, medium, and high workload score quantiles.

Alternative Hypothesis (H_1)

At least one workload quantile group has a significantly different calorie intake.

IMPORTANT

This hypothesis is based on **Insight 6**, where calorie intake appeared to increase and become more variable with higher workload score quantiles. Given the non-normality confirmed in HT2, the **Kruskal–Wallis test** was selected.

Results

- **Kruskal–Wallis Test Statistic:** $h =$ (from model output)
- **P-value:** $p < 0.05$

Conclusion

H_0 is rejected.

There is a statistically significant difference in calorie intake across workload score quantiles. In particular, higher workload intensity is associated with increased and more variable calorie consumption.

4.4. HT4: Weekday vs. Weekend Calorie Intake

Null Hypothesis (H_0)

There is no significant difference in calorie intake between weekdays and weekends.

Alternative Hypothesis (H_1)

There is a significant difference in calorie intake between weekdays and weekends.

IMPORTANT

Based on **EDA Insight 4**, workload scores were observed to be higher on weekdays, while calorie intake patterns differed between weekdays and weekends. Since two independent groups were compared and the calorie distribution was non-normal, the **Mann–Whitney U test** was applied.

Results

- **Mann–Whitney U Statistic:** $U =$ (from model output)
- **P-value:** $p < 0.05$

Conclusion

H_0 is rejected.

There is statistically significant evidence that calorie intake differs between weekdays and

weekends, highlighting the temporal dimension of the relationship between workload and dietary behavior.

5. Machine Learning Analysis

5.1 Objective

The objective of the machine learning analysis is to model and quantify the relationship between **academic workload intensity** and **daily calorie intake**, and to determine whether workload-related features can reliably predict caloric behavior.

Unlike purely predictive tasks, the primary focus of this section is **interpretability**—to identify which components of workload (lectures, homework, projects, and exams) most strongly influence calorie intake, and whether the relationship is linear or exhibits threshold-based behavior.

5.2 Feature Engineering

To capture both **daily stressors** and **cumulative academic pressure**, several engineered features were constructed prior to model training:

- **Daily Base Workload:**
A composite score reflecting same-day academic intensity derived from lecture attendance, homework, project, and exam indicators.
- **Weekly Fatigue Factor:**
A multiplicative factor accounting for accumulated workload across the week, reflecting prolonged stress exposure.
- **Workload Score:**
The final workload metric obtained by combining Daily Base Workload and Weekly Fatigue Factor.
- **Exam Week Indicator (Is_Exam_Week):**
A binary feature capturing sustained exam-related stress beyond single exam days.

All numerical features were standardized before training to ensure comparable scale and stable model performance.

5.3 Model Selection and Rationale

Given the behavioral nature of the problem and the relatively small dataset size, models were selected based on **interpretability**, **robustness**, and **ability to capture non-linear effects**.

The following models were evaluated:

Multiple Linear Regression

- Serves as a baseline interpretable model
- Quantifies direct linear effects between workload and calories
- Enables statistical inference through coefficient analysis

Decision Tree Regression

- Captures non-linear relationships and workload thresholds
- Particularly suitable for detecting abrupt increases in calorie intake

Random Forest Regression

- An ensemble approach used to validate feature importance consistency
- Reduces variance and increases robustness against noise

K-Means Clustering (Unsupervised)

- Used to explore natural groupings in workload–calorie behavior
- Provides structure validation independent of labeled outcomes

These models collectively balance explanatory power and behavioral insight.

5.4 Linear Regression Results

The multiple linear regression model indicates a **positive association between academic workload and calorie intake**, even after controlling for multiple workload components simultaneously.

Key observations include:

- **Exam-related variables** consistently exhibit the strongest positive coefficients
- The **exam week indicator** remains significant, suggesting sustained stress effects
- Lecture duration alone shows a weaker association compared to exams and combined workload

These results suggest that caloric behavior is influenced more by **psychological pressure** than by time spent in class.

5.5 Decision Tree Regression: Threshold Effects

The decision tree model reveals a **non-linear relationship** between workload and calorie intake.

A critical workload threshold emerges beyond which:

- Average daily calorie intake increases sharply
- Variability in caloric behavior becomes higher

This finding supports the hypothesis that **stress-related eating intensifies only after workload exceeds a certain level**, rather than increasing proportionally at all levels.

5.6 Random Forest Regression

The Random Forest model was used to validate feature importance rankings and improve robustness.

Feature importance analysis shows:

1. **Exam-related workload components**
2. **Weekly Fatigue Factor**
3. **Total Workload Score**

consistently rank above daily lecture load.

This confirms that **cumulative and high-pressure academic events**, rather than routine attendance, are the dominant drivers of caloric intake changes.

5.7 Unsupervised Learning: K-Means Clustering

K-Means clustering was applied using:

- Workload Score
- Daily calorie intake

Three distinct clusters emerged:

- **Low workload – stable calorie intake**
- **Moderate workload – moderate caloric increase**

- **High workload – elevated and volatile calorie intake**

The clustering results align closely with workload categories defined in the hypothesis testing stage, providing **unsupervised validation** of earlier findings.

6. Discussion

The machine learning results reinforce and extend the statistical findings:

- The workload–calorie relationship is **real but non-linear**
- Exam periods exert a **disproportionate effect** on eating behavior
- Calorie intake increases **before and during exam weeks**, not only on exam days

Importantly, models that capture **threshold behavior** outperform purely linear approaches in explaining dietary responses to stress.

7. Conclusion

This study demonstrates that academic workload is a meaningful predictor of daily calorie intake, particularly when accounting for **exam intensity** and **cumulative weekly stress**.

By integrating **custom workload modeling**, **statistical hypothesis testing**, and **interpretable machine learning techniques**, the analysis provides a robust explanation of stress-related eating behavior in an academic environment.

The findings highlight that:

- Stress effects accumulate over time
 - Behavioral responses intensify beyond critical workload thresholds
 - Exam-related pressure dominates routine academic activities
-

8. Suggestions for Future Work

Several extensions could further strengthen this research:

- Incorporating **macronutrient breakdowns** (protein, fat, carbohydrates)
- Applying **time-series models** (ARIMA, LSTM) to capture delayed effects

- Expanding the dataset to multiple individuals for population-level analysis
 - Integrating **sleep duration** and **physical activity** as mediating variables
-